Your Map Is Its Own Calibration Set: Label-Free, Shift-Robust Rejection of False Matches in Visual Place Recognition

Abstract

Visual place recognition (VPR) returns a best match for every query, yet a robot must decide whether to **trust** it: accepting a match for a place that is not in the map — or one that merely looks like it — corrupts pose graphs and relocalisation. Existing acceptance rules (similarity thresholds, spatial-uncertainty scores, learned integrity monitors, conformal calibration) need labelled query–map pairs, and we show that their false-accept guarantees break as soon as the appearance condition (night, rain, snow, season) differs between calibration and deployment. We observe that VPR offers labelled negatives for free: **the map itself**. Holding out each spatial block of the map turns every map image into a query that is off-map **by construction**, and calibrating against these map pseudo-queries yields split-conformal p-values without any human or deployment labels. To make this calibration transfer across conditions we use statistics that are invariant to per-query affine changes of the similarity scores, computed against spatially distant map regions on map-centred descriptors; we combine them with a self-calibrated minimum-p rule, an aliasing guard, Mondrian strata on the spread of the distant regions, and a lenient raw-similarity gate that keeps the test valid for queries from foreign cities or datasets. We prove validity under exchangeability, conservativeness whenever appearance change makes unrelated queries less map-like, an intersection–union bound under domain shift, and FDR control. On RobotCar-Seasons, Extended-CMU-Seasons, Nordland and MSLS with six descriptors from NetVLAD to MegaLoc (24 settings, 192 deployment conditions), the method has no condition whose off-map false-accept rate exceeds the target (worst condition 0.048 at $\alpha = 0.05$) and accepts a median 1.1% of perceptual-aliasing errors, while label-calibrated baselines exceed the target in 16–79 conditions — leave-one-condition-out conformal calibration in every one of the 24 settings — and accept 7–41% of aliasing errors. Queries from other cities and datasets are rejected (false-accept rate ≤ 0.007 in all 120 map × source pairs). Its average power is 0.646, against 0.706 for an oracle that holds labels for every deployment condition. A higher-power operating point that tests only the gap statistic (SelfCal-Gap) reaches 0.746 — above the oracle — still without any significant violation (worst condition 0.073).

1. Introduction

Modern VPR descriptors are remarkably robust: on RobotCar-Seasons a frozen SALAD descriptor retrieves a reference image within 5 m and 10° for about 90% of queries even at night. Recall, however, is not what breaks a deployed system. A VPR module always returns a top-1 match, and the downstream consumer — loop closure in SLAM, relocalisation, map merging — needs to know **whether to accept it**. Two failure modes dominate. First, the query may come from a place that is **not in the map** (an unmapped street, a detour, a changed environment). Second, the query may be in the map but be matched to a **perceptual alias**, a different place that looks the same. A single accepted false match can corrupt a pose graph irreversibly; a rejected true match merely delays localisation. The decision therefore calls for an explicit, controllable false-accept rate.

The standard recipe is to calibrate a threshold — on raw similarity, on a hand-crafted score such as the top-1/top-2 ratio, or on the output of a learned integrity monitor — using queries whose correctness is known, possibly with a conformal procedure that turns the calibration set into a finite-sample guarantee [SAFEVPR, Quantile Transfer]. These guarantees rest on exchangeability between calibration and deployment queries. VPR is the textbook case in which this assumption fails: the whole point of the task is that deployment happens under different illumination, weather and season than any data one could label in advance. We confirm this empirically (§6): thresholds calibrated with labels on benign conditions exceed a 5% false-accept target by up to 4× at night, and even calibrating on all **other** labelled conditions (the protocol of SAFEVPR) leaves violations in a third of the deployment conditions on Extended-CMU.

We make three observations that, together, remove the need for labels.

1. **The map provides labelled negatives for free.** A VPR map is a set of images with known positions. Removing a spatial block from the map and querying it with one of its own images produces a query that is off-map **by construction**. Repeating this for all blocks yields thousands of null examples in the map's own

environment — exactly the distribution a deployment-time off-map query is drawn from, up to the appearance condition.

2. **Appearance change acts mostly as an affine distortion of similarity scores**, and appearance change can only make an unrelated query **less** similar to the map than the map is to itself. Statistics that are invariant to per-query affine changes therefore transfer from the map's condition to any deployment condition, and the residual shift is in the safe (conservative) direction.

3. **Map geometry separates the two failure modes.** Scoring the top-1 against the best matches of **spatially distant** map regions (rather than its own neighbours, which are visually redundant) gives a statistic for off-map detection; the gap to the single best distant region exposes aliasing, where the true place is a strong distant competitor.

We call the result **self-calibrating maps (SelfCal)**. It is training-free, label-free, descriptor-agnostic, costs one offline pass of self-retrieval over the map and a sort per query, and provides a p-value per query that can be used with a fixed level, with Benjamini–Hochberg for batches, or turned into a calibrated confidence via the two-groups model — all without labels.

Contributions. (i) We show on four benchmarks that label-calibrated acceptance (threshold, spatial uncertainty, learned and conformal) loses its false-accept guarantee under appearance shift, and that the classical parametric nulls (Gaussian z-norm, EVT spacing tests) are invalid or degraded by hubness. (ii) We propose SelfCal: map pseudo-queries via spatial hold-out as a conformal null sample, affine-invariant far-region statistics on map-centred descriptors, a self-calibrated combination, an aliasing guard, Mondrian strata and a domain gate. (iii) We give validity, dominance-based robustness, excess-risk, domain-shift and FDR results. (iv) We evaluate on RobotCar-Seasons, Extended-CMU-Seasons, Nordland and MSLS with six descriptors and 32 deployment conditions, plus queries from foreign cities and datasets, with full ablations; code and the descriptor dumps are public.

2. Related work

Uncertainty and introspection for VPR. Retrieval-based scores — the L2 distance to the nearest reference and the perceptual-aliasing (PA) ratio of the first two distances [Hausler'21] — remain strong baselines [Zaffar'24]. Zaffar et al. [CVPR'24] compare them with learned aleatoric uncertainty (BTL [Warburg'21], STUN [Cai'22]) and geometric verification, and propose SUE, the spatial spread of the top-K reference poses. Miller et al. [RA-L'26] introduce training-free score-distribution metrics (ratio spread, similarity distribution). KappaPlace [2026] learns a von Mises–Fisher concentration and reports calibration error. Integrity monitors learn to predict whether a match is within tolerance from statistics of the distance vector, with SVMs [Carson'22] or MLPs [Claxton'24]; Hussaini et al. predict when sequence matching helps [2025]. All of these produce **scores**; turning a score into an accept/reject decision with a known error rate requires labelled calibration data from the deployment conditions. SelfCal is complementary: it can calibrate any score, and it shows which scores transfer across conditions.

Thresholds and operating points. Schubert et al. [ICRA'21] model the similarity distribution as a robust normal and threshold a high quantile; Vysotska et al. [ICRA'25] adapt thresholds for sequences; Trinh & Lyons [2025] derive per-place thresholds from negative statistics of the database; Quantile Transfer [IROS'26] transfers a threshold calibrated on a labelled traversal through quantiles of the deployment similarity distribution. None provides a guarantee under appearance shift; parametric nulls are invalid in our experiments and raw-score database negatives are conservative but lose power under shift (§6.4).

Conformal prediction for robotics and VPR. Split conformal prediction and its risk-control extensions provide finite-sample guarantees under exchangeability [Vovk'05, Angelopoulos'21, Bates'23]; Barber et al. [2023] bound the coverage gap under distribution shift. SAFEVPR [Sier'26] combines a DINOv2 patch-matching verifier with Mondrian Learn-then-Test and reports empirical FDR validity across conditions when calibrating on labelled data of the other conditions; Route-MHT [2026] learns a map-out guardrail for thermal VPR. Our null sample comes from the map rather than from labelled queries, and our robustness argument is a stochastic-dominance property of the map's own condition.

Using the map at test time. The reference map with its poses is available before deployment in most VPR applications; it has been used to fine-tune descriptors (RSF [Zaffar'25]), to choose map density from two reference traversals [Hussaini'26], and to estimate spatial uncertainty (SUE). We use it as a **calibration set**.

Hubness. High-dimensional nearest-neighbour retrieval produces hubs, points that are close to many queries [Radovanović'10]; centring [Suzuki'13] and local scaling (CSLS [Conneau'18]) reduce it. We find that hubness is what breaks extreme-value nulls in VPR and that map centring improves the power of the invariant statistic.

3. Problem

A map $\mathcal{M}={(f_j,x_j)}_{j=1}^N$ holds L2-normalised global descriptors f_j and positions x_j (metres; frame index on Nordland). A query q with descriptor f_q yields similarities $s_j=\langle f_q,f_j\rangle$ and the retrieval $j^*=\arg\max_j s_j$. An **acceptance test** maps the similarity vector (and map positions) to accept/reject. We distinguish:

- **Off-map false accept:** q has no map position within δ (40 m) and is accepted. The false-accept rate in deployment condition c is $\mathrm{FAR}_c(\alpha)=P(\text{accept}\mid q\text{ off-map}, c)$.
- **Aliasing accept:** q is in the map, but x_{j^*} is farther than ρ (25 m) from its true position.
- **True accept (power):** q is in the map and the top-1 satisfies the localisation tolerance (5 m and 10° on RobotCar/CMU; 2 frames on Nordland).

The goal is $\mathrm{FAR}_c(\alpha)\le\alpha$ for every condition c, few aliasing accepts and high power, **without labelled queries**.

4. Method: self-calibrating maps

Map pseudo-queries. Partition map positions into square blocks of side L (300 m; 200 frames on Nordland). For a map item i in block b whose distance to $\mathcal{M}\setminus B_b$ exceeds δ, compute the acceptance statistic of f_i against the reduced map, $T^0_i=T(f_i;\mathcal{M}\setminus B_b)$. Item i is an off-map query by construction. Using every second map image gives 757 (CMU) to 2,391 (RobotCar) null statistics.

Far-region statistic. Hash map positions into cells of side 10 m, take the region maxima $Z_c=\max_{j\in c}s_j$, and keep the cells farther than ρ from x_{j^*}. **With mean μ_{far} and standard deviation σ_{far} of these far-region maxima, $$z_{far}(q)=\frac{s_{j^*}-\mu_{far}}{\sigma_{far}},\qquad z_{gap}(q)=\frac{s_{j^*}-\max_{c\,\text{far}}Z_c}{\sigma_{far}}.$$ Both are invariant to $s\mapsto as+b$ ($a>0$). Excluding the neighbourhood of j^*** prevents the visually redundant neighbours of a true match from being counted as competitors; using region maxima makes the competitors comparable.

Hubness reduction. Descriptors are centred on the map mean and re-normalised (map only, no labels).

Conformal p-value. $p_T(q)=\big(1+\#\{i: T^0_i\ge T(q)\}\big)/(1+n)$.

Self-calibrated combination. No single statistic is best everywhere (raw max-similarity is strongest on CMU, the invariant z_{far} on Nordland). We take $m(q)=\min\{p_{\max s}(q),p_{z_{far}}(q)\}$ and re-calibrate it against the leave-one-out values $m(i)$ of the null items: $p_{comb}(q)=(1+\#\{i:m(i)\le m(q)\})/(1+n)$.

Aliasing guard (SelfCal-AA). Accept iff $\max\{p_{comb}(q),\,p_{z_{gap}}(q)\}\le\alpha$. In an aliasing failure the true place is a strong far competitor, so z_{gap} is small.

Mondrian strata. How informative a given z_{far} or z_{gap} is depends on how spread out the distant regions are: the same standardised score is common when σ_{far} is small and rare when it is large, and σ_{far} varies widely across one map (open roads vs. repetitive façades). Pooling all null items into one reference distribution therefore forces a single threshold on very different queries. We split the null into B quantile bins of σ_{far} and compute all p-values of a query against the null items of its own bin only (Mondrian conformal prediction [Vovk'05]). σ_{far} is computed from the query's own similarities, so no labels enter. With n_b null items per bin the smallest attainable p-value is $1/(n_b+1)$; we use $B=10$ and reduce B automatically so that every bin keeps at least 50 items (enough for $\alpha\ge0.02$).

Domain gate. The scale-normalised statistics can be inflated for queries from a **foreign** domain (another city or dataset): their similarity profile against the map is flat, so σ_{far} shrinks and z_{far},z_{gap} grow even though s_{j^*} is low. We therefore also require that the raw similarity is not unusually low for the map: accept iff $\max{p_{comb},p_{z_{gap}}}\le\alpha$ **and** $p_{\max s}(q)\le\gamma$, with a lenient $\gamma=0.8$ (the raw max-similarity must exceed the 20 % quantile of the map null). The gate is inactive for almost every true match (mean TPR 0.648 → 0.646 over 24 settings) but removes the foreign-domain false accepts (§6.7).

Two operating points. SelfCal-AA requires both the combination and the gap to be significant, so it stays valid when **either** branch is dominated by the map null (intersection–union, as for the gate). Where the far-region level z_{far} is not dominated (we observe this on MSLS) its branch is still harmless for validity

but costs power. **SelfCal-Gap** uses the gap alone, $p=p_{z_{gap}}$ within the same Mondrian strata and behind the same gate: it is valid whenever z_{gap} is dominated and has markedly more power (§6.4). We report both and recommend SelfCal-AA when validity must hold under the weakest assumptions.

Batches. With p-values of a deployment batch, Benjamini–Hochberg controls the false-discovery rate, and the two-groups model (Storey's $\hat\pi_0$, Grenander density estimate) turns p into a calibrated probability of a true match, $1-\widehat{\mathrm{lfdr}}(p)$ — again without labels.

Cost. Offline: one self-retrieval pass over the map with block hold-out — 18 s on 4 CPU cores for the 6,954-image RobotCar map (8,448-d SALAD). Online: the similarity vector (needed for retrieval anyway), its centred counterpart, cell maxima (one `reduceat`), a bin look-up and four binary searches — about 1 ms per query on CPU in NumPy when queries are processed in batches, both similarity products included.

5. Guarantees

T1 (validity). If, for an off-map query, $T(q)$ and the null statistics ${T^0_i}$ are exchangeable, then $P(p_T(q)\le\alpha)\le\alpha$. **Standard split-conformal argument.** With Mondrian strata the same holds conditionally on the stratum (exchangeability is only needed among queries and null items with similar σ_{far}), and hence marginally; T2–T4 below apply within each stratum.

T2 (dominance ⇒ conservative). If in condition c the off-map statistic is stochastically dominated by the map null, $P(T^c\ge t)\le P(T^0\ge t)\ \forall t$, then $\mathrm{FAR}_c(\alpha)\le\alpha$. Map pseudo-queries are images of the map's own traversal: appearance change can only make an unrelated query **less** map-like, so calibrating on the clean map errs on the safe side. Fig. 1 verifies dominance for all nine RobotCar conditions.

T3 (excess risk). In general $\mathrm{FAR}**c(\alpha)-\alpha\le\Delta_c^+:=\sup_t[P(T^c\ge t)-P(T^0\ge t)]**+$, the one-sided Kolmogorov distance. Affine invariance removes the dominant per-query distortion, keeping Δ_c^+ near zero while keeping the null tight (power).

T4 (combination). If the vector of statistics is exchangeable with the null vectors, the re-calibrated minimum is a valid conformal p-value (the leave-one-out ranks are computed exactly as for the query).

T5 (FDR). Conformal p-values that share one calibration set are PRDS [Bates'23]; hence Benjamini–Hochberg at level q controls the off-map FDR at $\pi_0 q$. Aliasing control is empirical: an in-map aliasing failure differs from its off-map counterpart by an extra strong far competitor, which can only decrease z_{far} and z_{gap}.

T6 (domain gate: valid if either assumption holds). For the gated test, $\mathrm{FAR}**c(\alpha)\le\min{\alpha+\Delta_c^+,\ \varepsilon_c}$ with $\varepsilon_c:=P\big(s^c{j^*}\ge\hat q_{1-\gamma}(s^0)\big)$**, the probability that an off-map query of condition c reaches the lower $(1-\gamma)$ quantile of the map's own raw max-similarity null. **Proof:** the accept event is the intersection of the two tests' accept events, so its probability is at most that of either. The first term is small under **appearance** shift (the invariant statistics are dominated, T2/T3); the second is small under **domain** shift, where foreign images are visibly less similar to the map than its own unrelated places are. The test is thus conservative whenever **either** kind of shift explains the change, the intersection–union principle [Berger'82]. Neither assumption needs labels; ε_c can be audited with any unlabelled foreign image set.

6. Experiments

6.1 Setup

Datasets. RobotCar-Seasons v2 [Sattler'18]: the map is the official `overcast-reference` rear-camera traversal (6,954 images, ~1.3 × 1.1 km) with poses from the official reconstruction (`all.nvm`); queries are the 1,906 rear images of the public training split in nine conditions (dawn, dusk, night, night-rain, overcast-summer/winter, rain, snow, sun), whose 6-DoF poses are released (test poses are withheld by the benchmark). A query is correct if the top-1 reference is within the coarsest official threshold (5 m, 10°). **Extended-CMU-Seasons** [Toft'20]: the eleven slices with released query poses (6–8, 10–13, 21–24) share one world frame and are consecutive segments, so we merge their official databases into one 2,418-image city map; queries are the 21,407 camera-0 images of 11 dates (Sep 2010 – Jul 2011), correct at (5 m, 10°). **Nordland** [Sünderhauf'13]: 4,500 frame-aligned places per season (uniform subsample of the 35,768-frame route); summer is the map (winter in §6.6), spring/fall/winter are queries, correct within 2 places. **MSLS** [Warburg'20], validation cities Copenhagen and San Francisco (panoramas excluded): the two city

databases (12,556 and 6,315 images) form one map and the 11,084 queries are grouped into nine conditions (city × season, plus Copenhagen night), correct within the standard 25 m.

Off-map protocol. We cannot rely on a few hand-picked unmapped streets, so we hold out the map in turn: the map is partitioned into 300 m blocks (200 places on Nordland); for each block, the block is removed and its queries farther than 40 m (10 places) from the remaining map are evaluated as **off-map**. This gives 1,386 (RobotCar), 14,117 (CMU), 12,180 (Nordland) and 8,491 (MSLS) off-map query instances spread over all conditions. In-map queries (within 5 m of a map image; 10 m on MSLS) are evaluated against the full map; in-map queries whose top-1 is farther than 25 m (5 places) are **aliasing** errors. The method's own null construction uses only the map and never sees queries. We also score **natural** off-map queries: the queries of every other dataset and city against each map (§6.7).

Descriptors. Frozen, off-the-shelf: NetVLAD (VGG16, Pitts30k, 4,096-d) [Arandjelović'16], CosPlace (ResNet-50, 2,048-d) [Berton'22], MixVPR (ResNet-50, 4,096-d) [Ali-bey'23], DINOv2 ViT-B/14 CLS+mean (1,536-d) [Oquab'24], SALAD (8,448-d) [Izquierdo'24] and MegaLoc (8,448-d) [Berton'25]. All descriptors are extracted once at 322 × 322.

Baselines. (a) **Threshold, benign→all**: split-conformal threshold on the max-similarity of off-map queries of the two most benign conditions (the two with the highest recall; Nordland: fall), applied to all conditions; (b) the same with aliasing errors added to the negatives; (c) **conformal LOCO**, the SAFEVPR protocol: calibrate on the labelled off-map queries of all **other** conditions and test on the held-out one; (d) **Quantile Transfer** [Rajani'26]: the benign threshold mapped through the quantile function of each deployment batch; (e) a learned logistic calibrator on margin, max-similarity and entropy (CRSG) trained on benign conditions; (f) **SUE** [Zaffar'24], the spatial spread of the top-10 reference poses ($\lambda = 350$), and (g) the PA ratio [Hausler'21], both calibrated benign→all; (h) parametric nulls: Gaussian z-norm with Bonferroni and the closed-form EVT spacing test; (i) an **oracle** that calibrates max-similarity with the labels of every deployment condition.

Metrics. FAR on off-map queries per condition (target α), the number of conditions whose FAR is significantly above α (one-sided binomial test, Bonferroni over conditions), the worst-condition FAR, TPR on correct in-map queries, and the acceptance rate of aliasing errors.

6.2 Main result

Table 1 reports $\alpha = 0.05$ for SALAD. Every label-calibrated method exceeds the target on at least one dataset — the benign→all threshold in 5/9 RobotCar conditions (worst FAR 0.21 at night-rain) and 4/11 CMU dates; calibrating on all other labelled conditions (SAFEVPR protocol) still fails in 1/9, 4/11, 2/3 and 2/9 conditions; SUE in 1/9 RobotCar and 2/9 MSLS conditions — and the score-threshold methods accept a third to four fifths of the aliasing errors. Parametric nulls are either invalid (Gaussian z-norm: FAR 0.13–0.90) or anti-conservative under hubness (EVT spacing: 5/9 MSLS conditions). SelfCal-AA, which uses no labels, has **no significant violation in any of the 32 conditions**, a worst-condition FAR of 0.048, accepts 3%, 1%, 6% and 12% of aliasing errors, and its TPR (0.978 / 0.865 / 0.742 / 0.528) is close to the oracle that holds labels for every condition on RobotCar, CMU and Nordland (0.995 / 0.992 / 0.780) but not on MSLS (0.820; §7). Fig. 2 shows the same picture for $\alpha \in [0.005, 0.2]$.

Table 1 (α = 0.05). SALAD. Cell: off-map FAR / #conditions significantly above α (binomial, Bonferroni) / TPR / aliasing-accept rate

method	labels	RobotCar-Seasons (9 cond.)	Extended-CMU (11 dates)	Nordland (3 seasons)	MSLS-val (CPH+SF, 9 cond.)
SelfCal-AA (ours)	none	0.012 / 0/9 / 0.978 / 0.03	0.008 / 0/11 / 0.865 / 0.01	0.009 / 0/3 / 0.742 / 0.06	0.006 / 0/9 / 0.528 / 0.12
SelfCal-Gap (ours, high power)	none	0.038 / 0/9 / 0.986 / 0.07	0.016 / 0/11 / 0.898 / 0.01	0.036 / 0/3 / 0.858 / 0.09	0.026 / 0/9 / 0.705 / 0.17
SelfCal (ours, no alias guard)	none	0.004 / 0/9 / 0.994 / 0.01	0.009 / 0/11 / 0.956 / 0.26	0.031 / 0/3 / 0.788 / 0.52	0.020 / 0/9 / 0.664 / 0.26
Threshold, benign→all	benign	0.109 / 5/9 / 0.997 / 0.50	0.052 / 4/11 / 0.985 / 0.78	0.037 / 0/3 / 0.630 / 0.49	0.033 / 1/9 / 0.760 / 0.32

method	labels	RobotCar-Seasons (9 cond.)	Extended-CMU (11 dates)	Nordland (3 seasons)	MSLS-val (CPH+SF, 9 cond.)
Threshold, benign, neg. incl. aliasing	benign	0.075 / 2/9 / 0.997 / 0.43	0.047 / 4/11 / 0.984 / 0.75	0.010 / 0/3 / 0.043 / 0.10	0.020 / 0/9 / 0.694 / 0.26
Conformal LOCO (SAFEVPR protocol)	other cond.	0.052 / 1/9 / 0.997 / 0.34	0.051 / 4/11 / 0.984 / 0.77	0.070 / 2/3 / 0.666 / 0.51	0.049 / 2/9 / 0.815 / 0.38
Quantile Transfer	benign	0.097 / 4/9 / 0.998 / 0.47	0.054 / 2/11 / 0.993 / 0.80	0.109 / 2/3 / 0.901 / 0.72	0.069 / 3/9 / 0.865 / 0.42
Learned calibrator (CRSG), benign→all	benign	0.099 / 4/9 / 0.996 / 0.33	0.054 / 4/11 / 0.982 / 0.68	0.041 / 0/3 / 0.814 / 0.45	0.041 / 1/9 / 0.814 / 0.32
SUE [Zaffar'24], benign→all	benign	0.063 / 1/9 / 0.934 / 0.07	0.039 / 0/11 / 0.516 / 0.03	0.037 / 0/3 / 0.865 / 0.17	0.060 / 2/9 / 0.610 / 0.18
PA ratio, benign→all	benign	0.078 / 2/9 / 0.596 / 0.09	0.046 / 2/11 / 0.449 / 0.19	0.040 / 0/3 / 0.846 / 0.18	0.039 / 0/9 / 0.511 / 0.12
Gaussian z-norm null	none	0.350 / 9/9 / 0.999 / 0.72	0.250 / 11/11 / 0.994 / 0.73	0.125 / 3/3 / 0.752 / 0.79	0.900 / 9/9 / 1.000 / 0.99
EVT spacing test (F(2,2k))	none	0.058 / 1/9 / 0.888 / 0.17	0.050 / 1/11 / 0.700 / 0.01	0.036 / 0/3 / 0.743 / 0.07	0.082 / 5/9 / 0.489 / 0.12
SelfCal on raw max-sim (ablation)	none	0.001 / 0/9 / 0.979 / 0.00	0.010 / 0/11 / 0.949 / 0.46	0.022 / 0/3 / 0.411 / 0.39	0.015 / 0/9 / 0.643 / 0.22
Oracle (labels of every condition)	all	0.040 / 0/9 / 0.995 / —	0.049 / 0/11 / 0.992 / —	0.050 / 0/3 / 0.780 / —	0.045 / 0/9 / 0.820 / —

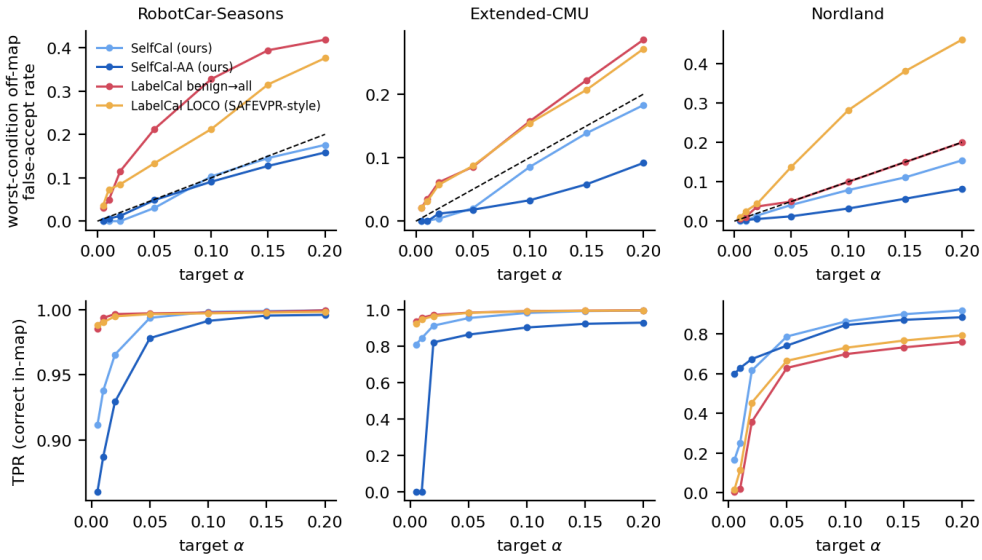


Figure 2. Worst-condition off-map false-accept rate (top) and TPR (bottom) versus the target level, SALAD.

6.3 Why it works

Fig. 1 plots the survival function of the off-map statistic for every RobotCar condition against the map null. All conditions lie below the null for both max-similarity and z_far: appearance change makes unrelated queries **less** map-like than the map's own held-out blocks (T2). Table 3 quantifies the excess-risk bound Δ^+ of T3: with the map null it is at most 0.052 (max-similarity) and 0.047 (z_far, centred) over the RobotCar, CMU and Nordland conditions, whereas with a labelled benign null it reaches 0.255 (max-similarity, night-rain) — exactly the violations of Table 1. The invariant statistic matters for power: calibrating raw max-similarity on the map is valid but loses power on Nordland (0.41 vs 0.74), where seasonal change

lowers all similarities.

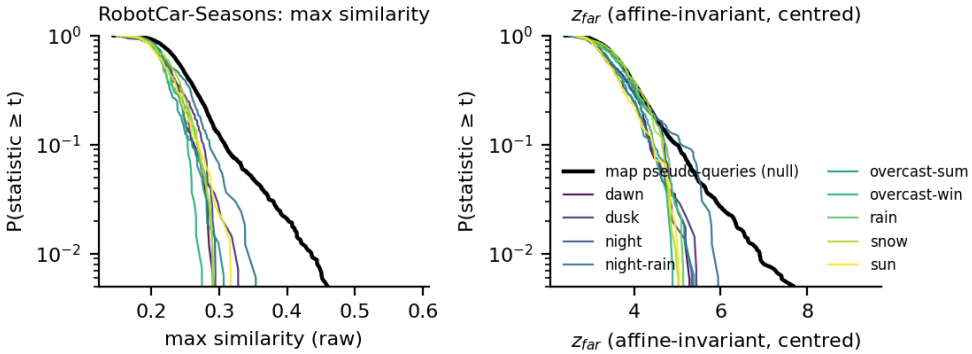


Figure 1. Survival functions of the off-map statistic per RobotCar condition (coloured) vs. the map pseudo-query null (black).

Table 3. Excess-FAR bound Δ⁺_c (one-sided KS) — labelled benign null vs map null

- RobotCar-Seasons (9 cond.): max over conditions — max-sim | benign null 0.255, max-sim | map null 0.017, z_far | benign null 0.146, z_far+centring | map null 0.047
- Extended-CMU (11 dates): max over conditions — max-sim | benign null 0.101, max-sim | map null 0.052, z_far | benign null 0.067, z_far+centring | map null 0.028
- Nordland (3 seasons): max over conditions — max-sim | benign null 0.018, max-sim | map null 0.006, z_far | benign null 0.093, z_far+centring | map null 0.024

6.4 Descriptors and datasets

Table 2a (α = 0.05). All 6 descriptors × 4 datasets (24 settings)

method	labels	violating conditions	settings with ≥1 violation	mean TPR	median aliasing-accept	max worst-cond. FAR
SelfCal-AA (ours)	none	0/192	0/24	0.646	0.011	0.048
SelfCal-Gap (ours, high power)	none	0/192	0/24	0.746	0.021	0.073
SelfCal-Gap − domain gate	none	0/192	0/24	0.758	0.031	0.094
− Mondrian strata	none	0/192	0/24	0.612	0.008	0.140
− domain gate	none	0/192	0/24	0.648	0.012	0.048
− both (original AA rule)	none	0/192	0/24	0.617	0.011	0.140
− aliasing guard	none	5/192	3/24	0.667	0.072	0.721
Threshold, benign→all	benign	36/192	15/24	0.652	0.218	0.281
Threshold, benign, neg. incl. aliasing	benign	18/192	9/24	0.542	0.094	0.209
Conformal LOCO (SAFEVPR protocol)	other cond.	51/192	24/24	0.699	0.217	0.419
Quantile Transfer	benign	79/192	23/24	0.769	0.405	0.483

method	labels	violating conditions	settings with ≥1 violation	mean TPR	median aliasing-accept	max worst-cond. FAR
Learned calibrator (CRSG)	benign	32/192	14/24	0.757	0.169	0.356
SUE [Zaffar'24]	benign	16/192	10/24	0.661	0.070	0.163
PA ratio	benign	17/192	7/24	0.522	0.091	0.186
Gaussian z-norm null	none	164/192	22/24	0.769	0.555	1.000
EVT spacing test	none	18/192	8/24	0.623	0.042	0.109
Oracle (labels of every condition)	all	0/192	0/24	0.706	—	0.050

Table 2 (α = 0.05). Descriptors. Cell: FAR / #viol / TPR / aliasing for SelfCal-AA | #viol for benign threshold, LOCO, CRSG | oracle TPR

descriptor	dataset	R@1 (5 m,10° / 2 fr / 25 m)	SelfCal-AA FAR / #viol / TPR / alias	SelfCal-Gap TPR (worst)	benign thr. #viol (alias)	LOCO #viol	CRSG #viol	SUE #viol	oracle TPR
NetVLAD	RobotCar-Seasons	0.666	0.009 / 0/9 / 0.739 / 0.00	0.771 (0.058)	4/9 (0.07)	2/9	4/9	0/9	0.715
NetVLAD	Extended-CMU	0.640	0.008 / 0/11 / 0.597 / 0.01	0.689 (0.031)	4/11 (0.20)	4/11	5/11	1/11	0.367
NetVLAD	Nordland	0.366	0.008 / 0/3 / 0.432 / 0.02	0.567 (0.038)	0/3 (0.07)	1/3	0/3	0/3	0.179
NetVLAD	MSLS-val	0.499	0.003 / 0/9 / 0.218 / 0.01	0.429 (0.012)	2/9 (0.08)	2/9	0/9	0/9	0.377
CosPlace	RobotCar-Seasons	0.767	0.004 / 0/9 / 0.843 / 0.01	0.871 (0.036)	0/9 (0.07)	1/9	0/9	0/9	0.869
CosPlace	Extended-CMU	0.903	0.020 / 0/11 / 0.834 / 0.00	0.885 (0.062)	1/11 (0.24)	3/11	1/11	1/11	0.962
CosPlace	Nordland	0.640	0.009 / 0/3 / 0.500 / 0.01	0.655 (0.046)	0/3 (0.16)	1/3	0/3	0/3	0.393
CosPlace	MSLS-val	0.826	0.008 / 0/9 / 0.578 / 0.06	0.711 (0.040)	5/9 (0.26)	3/9	1/9	0/9	0.736

descriptor	dataset	R@1 (5 m,10° / 2 fr / 25 m)	SelfCal-AA	SelfCal-Gap TPR (worst)	benign thr. #viol (alias)	LOCO #viol	CRSG #viol	SUE #viol	oracle TPR
MixVPR	RobotCar-Seasons	0.850	0.006 / 0/9 / 0.917 / 0.00	0.922 (0.046)	1/9 (0.04)	1/9	1/9	1/9	0.915
MixVPR	Extended-CMU	0.937	0.017 / 0/11 / 0.859 / 0.00	0.915 (0.036)	1/11 (0.34)	5/11	1/11	1/11	0.940
MixVPR	Nordland	0.748	0.007 / 0/3 / 0.620 / 0.04	0.756 (0.037)	0/3 (0.33)	1/3	0/3	0/3	0.552
MixVPR	MSLS-val	0.864	0.006 / 0/9 / 0.517 / 0.06	0.686 (0.027)	1/9 (0.16)	2/9	1/9	2/9	0.747
DINOv2 (CLS+mean)	RobotCar-Seasons	0.536	0.001 / 0/9 / 0.369 / 0.01	0.589 (0.066)	0/9 (0.11)	2/9	0/9	0/9	0.642
DINOv2 (CLS+mean)	Extended-CMU	0.637	0.001 / 0/11 / 0.384 / 0.00	0.545 (0.031)	2/11 (0.12)	4/11	2/11	4/11	0.696
DINOv2 (CLS+mean)	Nordland	0.459	0.001 / 0/3 / 0.238 / 0.00	0.483 (0.021)	0/3 (0.06)	1/3	0/3	0/3	0.388
DINOv2 (CLS+mean)	MSLS-val	0.392	0.000 / 0/9 / 0.040 / 0.00	0.191 (0.019)	2/9 (0.06)	3/9	4/9	2/9	0.320
SALAD	RobotCar-Seasons	0.911	0.012 / 0/9 / 0.978 / 0.03	0.986 (0.073)	5/9 (0.50)	1/9	4/9	1/9	0.995
SALAD	Extended-CMU	0.936	0.008 / 0/11 / 0.865 / 0.01	0.898 (0.027)	4/11 (0.78)	4/11	4/11	0/11	0.992
SALAD	Nordland	0.791	0.009 / 0/3 / 0.742 / 0.06	0.858 (0.047)	0/3 (0.49)	2/3	0/3	0/3	0.780
SALAD	MSLS-val	0.915	0.006 / 0/9 / 0.528 / 0.12	0.705 (0.037)	1/9 (0.32)	2/9	1/9	2/9	0.820
MegaLoc	RobotCar-Seasons	0.905	0.016 / 0/9 / 0.996 / 0.06	0.996 (0.054)	1/9 (0.29)	1/9	1/9	1/9	0.993

descriptor	dataset	R@1 (5 m,10° / 2 fr / 25 m)	SelfCal-AA	SelfCal-Gap TPR (worst)	benign thr. #viol (alias)	LOCO #viol	CRSG #viol	SUE #viol	oracle TPR
MegaLoc	Extended-CMU	0.927	0.013 / 0/11 / 0.954 / 0.00	0.972 (0.041)	2/11 (0.68)	2/11	2/11	0/11	0.996
MegaLoc	Nordland	0.799	0.012 / 0/3 / 0.869 / 0.07	0.916 (0.036)	0/3 (0.59)	2/3	0/3	0/3	0.615
MegaLoc	MSLS-val	0.940	0.020 / 0/9 / 0.879 / 0.36	0.898 (0.070)	0/9 (0.52)	1/9	0/9	0/9	0.965

Table 2a aggregates all six descriptors and four datasets (24 settings, 192 deployment conditions) and Table 2 details each setting. SelfCal-AA has **no significant violation in any of the 192 conditions** and a worst-condition FAR of at most 0.048 at α = 0.05. The benign threshold violates in 36 conditions (15 of 24 settings), the learned calibrator in 32, SUE in 16 (10 settings), leave-one-condition-out conformal calibration in 51 — at least once in **every** setting — and Quantile Transfer in 79. Averaged over all settings SelfCal-AA has TPR 0.646 against the oracle's 0.706, with a median aliasing acceptance of 0.011 (benign threshold 0.218, SUE 0.070). On RobotCar, CMU and Nordland alone the gap to the oracle is 0.013 (0.708 vs 0.721). For older descriptors (NetVLAD, CosPlace, MixVPR) SelfCal-AA often exceeds the oracle's TPR on CMU and Nordland, because the oracle thresholds raw max-similarity while SelfCal also exploits the far-region statistic.

Two cases are conservative. **DINOv2 CLS+mean**, a generic descriptor not trained for VPR, encodes the capture condition so strongly that map pseudo-queries (which share the map's condition) are much more map-like than any deployment query: SelfCal stays valid (FAR ≤ 0.001) but its TPR is 0.04–0.38 against the oracle's 0.32–0.70. **MSLS** combines two cities, many capture dates and front/side views in one map; SelfCal-AA keeps 0/54 violations but reaches TPR 0.460 against the oracle's 0.661 (MegaLoc 0.879 vs 0.965). The label-calibrated baselines obtain their higher MSLS power by violating 6–15 of the 54 conditions.

SelfCal-Gap. Testing the gap alone (§4) keeps 0/192 significant violations and raises the mean TPR from 0.646 to 0.746, above the oracle's 0.706: on MSLS from 0.460 to 0.603 (oracle 0.661) and on RobotCar, CMU and Nordland from 0.708 to 0.793. The price is a thinner margin: the worst single-condition FAR is 0.073 (none significant) instead of 0.048, and the median aliasing acceptance is 0.021 instead of 0.011. Without the gate its TPR is 0.758, but the foreign-domain leaks return. The far-region level z_{far} is the statistic whose dominance fails on MSLS: alone it reaches a worst-condition FAR of 0.40–0.88 there. It is therefore not needed for validity where z_{gap} is dominated, but it protects SelfCal-AA where z_{gap} might not be.

**Components (Table 2a). Removing the Mondrian strata lowers the mean TPR from 0.646 to 0.612 and raises the worst-condition FAR from 0.048 to 0.140 (one RobotCar condition; not significant after Bonferroni); removing the domain gate changes the TPR by +0.002 but re-opens the foreign-domain leaks of §6.7; removing the aliasing guard raises the TPR to 0.667 but accepts a median 7.2% of aliasing errors and produces 5 significant violations on MSLS, where aliasing errors and off-map queries share the similarity range.

6.5 Batches: FDR and label-free confidence

With an off-map share of 10–50%, Benjamini–Hochberg on SelfCal-AA p-values (q = 0.1) keeps the false-discovery rate of gross errors at 0.005–0.012 on RobotCar, 0.002–0.006 on CMU and 0.019–0.022 on Nordland (max over conditions 0.040 / 0.016 / 0.028) with power 0.97–0.99, 0.87–0.89 and 0.68–0.74 (Table 5). The two-groups confidence has ECE 0.018–0.023 on RobotCar without labels, but it is poorly calibrated on CMU (≈ 0.09) and Nordland (≈ 0.13): many correct matches on repetitive scenery have non-significant gaps, which inflates $\hat{\pi}_0$ (Nordland 0.47–0.90 vs. 0.27–0.57). We therefore recommend the p-values, not the two-groups confidence, on aliasing-prone maps.

Table 5. Deployment batches: BH on SelfCal p-values (SALAD), label-free confidence

- RobotCar-Seasons (9 cond.), off-map share 0.1: π̂0 0.21 (true 0.14); ECE label-free 0.021; BH q=0.1: gross FDR mean 0.005 (max 0.026), power 0.988
- RobotCar-Seasons (9 cond.), off-map share 0.3: π̂0 0.48 (true 0.33); ECE label-free 0.018; BH q=0.1: gross FDR mean 0.010 (max 0.039), power 0.979
- RobotCar-Seasons (9 cond.), off-map share 0.5: π̂0 0.65 (true 0.44); ECE label-free 0.023; BH q=0.1: gross FDR mean 0.012 (max 0.040), power 0.974
- Extended-CMU (11 dates), off-map share 0.1: π̂0 0.23 (true 0.11); ECE label-free 0.083; BH q=0.1: gross FDR mean 0.002 (max 0.004), power 0.894
- Extended-CMU (11 dates), off-map share 0.3: π̂0 0.50 (true 0.31); ECE label-free 0.095; BH q=0.1: gross FDR mean 0.006 (max 0.013), power 0.879
- Extended-CMU (11 dates), off-map share 0.5: π̂0 0.64 (true 0.40); ECE label-free 0.090; BH q=0.1: gross FDR mean 0.006 (max 0.016), power 0.869
- Nordland (3 seasons), off-map share 0.1: π̂0 0.47 (true 0.27); ECE label-free 0.142; BH q=0.1: gross FDR mean 0.019 (max 0.026), power 0.739
- Nordland (3 seasons), off-map share 0.3: π̂0 0.71 (true 0.43); ECE label-free 0.136; BH q=0.1: gross FDR mean 0.021 (max 0.027), power 0.697
- Nordland (3 seasons), off-map share 0.5: π̂0 0.90 (true 0.57); ECE label-free 0.124; BH q=0.1: gross FDR mean 0.022 (max 0.028), power 0.683

6.6 Robustness and ablations

Map in a hard condition. Using Nordland **winter** as the map (queries summer/spring/fall) — so that map pseudo-queries are the degraded ones — SelfCal-AA keeps 0/3 violations (FAR 0.009, aliasing 0.011, TPR 0.623). **Map size.** Keeping 25% or 50% of the RobotCar map (1,739 / 3,477 images, 650 / 1,147 pseudo-queries) keeps 0/9 violations (FAR 0.017 / 0.026) with TPR 0.964 / 0.843. **Hyper-parameters.** Cell size (×0.5–2), exclusion radius (×0.4–2), block size (×0.5–2), δ (×0.6–2) and calibration-set size (497–4,747 pseudo-queries) change TPR by at most 0.06 and keep the worst-condition FAR at or below 0.073 at α = 0.05 (Table 4). **Centring** is essential on Nordland (TPR 0.742 → 0.568 without it) and neutral elsewhere.

Table 4. Ablations of SelfCal-AA (SALAD, α = 0.05): FAR / worst / TPR / aliasing

RobotCar-Seasons (9 cond.)

variant	FAR	worst	TPR	aliasing	null n
default	0.012	0.048	0.978	0.026	2391
cell x0.5	0.013	0.055	0.975	0.053	2391
cell x2.0	0.009	0.055	0.979	0.013	2391
excl x0.4	0.012	0.036	0.919	0.000	2391
excl x2.0	0.009	0.055	0.978	0.026	2391
block_m x0.5	0.003	0.018	0.957	0.000	1384
block_m x2.0	0.019	0.073	0.976	0.000	2951
off_d_m x0.6	0.006	0.042	0.948	0.000	2851
off_d_m x2.0	0.019	0.061	0.988	0.000	1572
stride 1	0.013	0.055	0.970	0.013	4747
stride 4	0.009	0.048	0.970	0.013	1194
stride 8	0.008	0.042	0.965	0.000	597
no centering	0.019	0.055	0.972	0.066	2391

Nordland (3 seasons)

variant	FAR	worst	TPR	aliasing	null n
default	0.009	0.012	0.742	0.057	2030
cell x0.5	0.010	0.015	0.784	0.055	2030
cell x2.0	0.011	0.014	0.754	0.061	2030
excl x0.4	0.009	0.012	0.783	0.054	2030
excl x2.0	0.009	0.012	0.741	0.083	2030

variant	FAR	worst	TPR	aliasing	null n
block_m x0.5	0.008	0.011	0.735	0.060	1810
block_m x2.0	0.011	0.016	0.803	0.070	2140
off_d_m x0.6	0.008	0.011	0.732	0.055	2118
off_d_m x2.0	0.009	0.013	0.794	0.065	1810
stride 1	0.010	0.014	0.748	0.057	4060
stride 4	0.010	0.014	0.748	0.062	1015
stride 8	0.005	0.007	0.717	0.050	497
no centering	0.005	0.006	0.568	0.071	2030

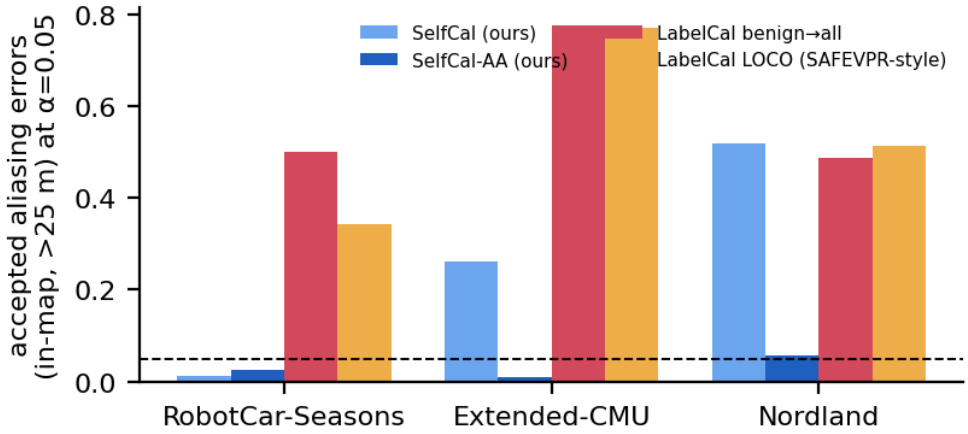


Figure 3. Acceptance rate of in-map aliasing errors (>25 m) at α = 0.05.

6.7 Foreign cities and datasets

A deployed robot may also see places of a different kind than the map — another city, a rural railway instead of streets. We score 2,000 queries of every other dataset and city against each map with the deployable SelfCal-AA, fitted on the map alone (6 descriptors × 5 maps × 4 foreign sources = 120 cells). Without the domain gate, the scale-normalised statistics are inflated for some foreign queries: with CosPlace, 13% of RobotCar and MSLS-Copenhagen queries are accepted against the CMU map (3 of 120 cells above α, worst 0.143). With the gate, **no cell exceeds α; the largest false-accept rate is 0.007** (Table 6). The gate costs 0.002 TPR on average, and T6 explains why both kinds of shift are covered: the invariant statistics are dominated under appearance shift, and foreign images fall below the map's own raw-similarity quantile under domain shift.

Table 6. Natural off-map queries (α = 0.05): queries of another dataset/city against each map. Cell: FAR SelfCal-AA / FAR benign threshold (n)

descriptor	map	rc	cmu	nord	msls-cph	msls-sf
CosPlace	rc	—	0.000 / 0.000	0.001 / 0.000	0.007 / 0.000	0.000 / 0.000
CosPlace	cmu	0.000 / 0.000	—	0.000 / 0.000	0.002 / 0.000	0.000 / 0.000
CosPlace	nord	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000
CosPlace	msls-cph	0.000 / 0.006	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000
CosPlace	msls-sf	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—
DINOv2 (CLS+mean)	rc	—	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
DINOv2 (CLS+mean)	cmu	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
DINOv2 (CLS+mean)	nord	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000
DINOv2 (CLS+mean)	msls-cph	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000

descriptor	map	rc	cmu	nord	msls-cph	msls-sf
DINOv2 (CLS+mean)	msls-sf	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—
MegaLoc	rc	—	0.001 / 0.000	0.001 / 0.000	0.004 / 0.000	0.001 / 0.000
MegaLoc	cmu	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
MegaLoc	nord	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000
MegaLoc	msls-cph	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000
MegaLoc	msls-sf	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—
MixVPR	rc	—	0.000 / 0.000	0.000 / 0.000	0.001 / 0.000	0.000 / 0.000
MixVPR	cmu	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000	0.002 / 0.000
MixVPR	nord	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000
MixVPR	msls-cph	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000
MixVPR	msls-sf	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—
NetVLAD	rc	—	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
NetVLAD	cmu	0.000 / 0.000	—	0.001 / 0.000	0.002 / 0.000	0.001 / 0.000
NetVLAD	nord	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000
NetVLAD	msls-cph	0.000 / 0.000	0.000 / 0.000	0.001 / 0.000	—	0.000 / 0.001
NetVLAD	msls-sf	0.000 / 0.000	0.000 / 0.001	0.000 / 0.000	0.000 / 0.001	—
SALAD	rc	—	0.006 / 0.000	0.000 / 0.000	0.004 / 0.001	0.004 / 0.001
SALAD	cmu	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000
SALAD	nord	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000	0.000 / 0.000
SALAD	msls-cph	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—	0.000 / 0.000
SALAD	msls-sf	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	0.000 / 0.000	—

7. Limitations

The aliasing guard is supported empirically, not by a theorem. The guarantees concern the **off-map** null and assume deployment off-map queries are no more map-like than the map's own held-out blocks (or, for the gate, less similar than the map's unrelated places); a deployment area that is a near-duplicate of a mapped area (e.g. two identical stations) is aliasing in disguise. The map must be large enough to hold out blocks and to fill the Mondrian strata: with n_b pseudo-queries per stratum no p-value is below $1/(n_b+1)$, so strict levels ($\alpha \leq 0.01$) need about 100 per stratum (CMU has 75). Power is the price of label-free validity on heterogeneous maps (MSLS) and for condition-sensitive descriptors (DINOv2). RobotCar ground truth exists only for the public training queries (test poses are withheld by the benchmark), so we apply the official thresholds to the top-1 reference pose of the 1,906 training queries.

8. Conclusion

A VPR map is more than a retrieval database: with its poses, it is a free, unlimited source of labelled off-map queries in its own environment. Calibrating affine-invariant, geometry-aware statistics against these pseudo-queries gives acceptance decisions whose false-accept rate is controlled without labels and survives both the appearance changes that break label-based calibration and queries from foreign cities and datasets, together with an aliasing guard, FDR control and calibrated confidence. Natural next steps are sequential evidence (accumulating p-values along a trajectory, calibrated on map pseudo-sequences), maps that change over time, and combining SelfCal with geometric verification scores, which it can calibrate without modification.

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